

SPIKE GATE PHASE 2 METHODOLOGY

Component-constrained optimisation of SEP and MTOjects on science images

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METHODOLOGICAL PRINCIPLE

Spike Gate supplies evidence from residual images; SEP and MTOjects segment only the original science images. Gate evidence selects credible connected components, but never becomes the science-image input.

1. PURPOSE AND RATIONALE

Earlier objective-only agreement could reward a large mask simply because it overlapped a Spike Gate target. Phase 2 changes the gate from a score-only term into an explicit connected-component selector. Recovery credit is conditional on compact, gate-supported science-image segmentation.

2. PROCESSING ARCHITECTURE

1. Generate Spike Gate candidate evidence from the residual image.
2. Run SEP or MTOjects on the original science image within documentation-supported parameter ranges.
3. Label connected components in native science-image coordinates.
4. Project each native label map once into the common deprojected diagnostic view.
5. Retain only components satisfying target intersection, support fraction and size constraints.
6. Apply the accepted mask and replace masked profile samples with a log-linear bridge.

3. COMPONENT RETENTION RULES

Target intersection: the component must intersect the compact gate target.

Support fraction: at least 20% of projected component pixels must fall inside permitted support.

Maximum extent: a component may occupy no more than 8% of the diagnostic view.

Coordinate handling: native labels are projected once to avoid denominator drift.

4. OPTIMISATION OBJECTIVE

The scalar objective balances gate recovery, candidate detection, supported-mask precision, excess area, protected galaxy loss, native masked fraction, non-gate profile masking, bridge span and an explicit zero-detection penalty when credible candidates exist.

5. AUDIT AND RELEASE GATE

Fifteen gate-positive, high-risk galaxies formed a stress-test audit. Each algorithm used a short 24-trial search. MTO was permitted to proceed as a conservative diagnostic run; SEP was not.

6. IMPLEMENTATION

Objective: Foreground Masking/Shared/spike_gate_objective.py

Batch filter: Foreground Masking/Shared/spike_gate_component_filter.py

SEP optimiser: Foreground Masking/Optimisation/optimize_spike_gate_SEP.py

MTO optimiser: Foreground Masking/Optimisation/optimize_spike_gate_MTOObjects.py

Audit evaluator: Foreground Masking/Optimisation/evaluate_constrained_spike_gate_batch.py

7. INTERPRETATION LIMITS

Low masked area demonstrates conservatism, not necessarily correct foreground recovery. Phase 2 is a profile-protection diagnostic and is not yet a complete foreground-object catalogue.